

- 5 The table shows the energy content in the three trophic levels of a food chain from an African plain.

Trophic level	Organism	Energy content
3	Lion 	920
2	Gerenuk 	8 825
1	Tree 	90 480

- (a) (i) Which are the units for the energy content of these organisms?

(1)

- ☐ **A** $\text{kJ m}^{-1} \text{ year}^{-1}$
- ☐ **B** $\text{kJ m}^{-1} \text{ year}^{-2}$
- ☐ **C** $\text{kJ m}^{-2} \text{ year}^{-1}$
- ☐ **D** $\text{kJ m}^{-2} \text{ year}^{-2}$

(ii) The energy content of the tree is the net primary productivity (NPP).

What is the relationship between NPP, gross primary productivity (GPP) and respiration (R)?

(1)

☐ **A** $NPP = GPP + R$

☐ **B** $NPP = GPP - R$

☐ **C** $NPP = GPP \times R$

☐ **D** $NPP = GPP \div R$

(iii) Calculate the ratio of the energy contents shown in these trophic levels:

trophic level 1 : trophic level 2 : trophic level 3

(1)

Answer

(iv) Explain why there is no fourth trophic level in this food chain.

(2)

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(v) Describe the role of microorganisms in recycling carbon present in this food chain.

(3)

*(b) The photographs show gerenuk and a springbok to the same magnification.



gerenuk



(Source: © David Havel / Alamy Stock Photo)

springbok

The table gives some information about gerenuk and springbok.

Information	Gerenuk	Springbok
Diet	leaves from a range of different trees, often spiny trees, up to 2 m from the ground	short grass and bushes
Water requirements	met by water content of food	will drink water when available
Height / cm	80 to 105	69 to 87
Mass / kg	male 45 female 30	male 41 female 37
Colour of fur	light brown with a white underbelly	light to mid brown with large areas of white
Length of horns / cm	25 to 44 in males only	35 to 49 in both males and females but narrower horns in females

(Total for Question 5 = 14 marks)